

# SOLIDWORKS DimXpert API (.NET) — Reference

## (6/8): IDimXpertIntersectPointFeature- IDimXpertPatternFeature

SOLIDWORKS API Documentation (converted from CHM)

# IDimXpertIntersectPointFeature (Type)

**IDimXpertIntersectPointFeature Interface**  
[See Also](#) [Members](#) [Example](#)

 

SolidWorks.Interop.swdimxpert Namespace : IDimXpertIntersectPointFeature Interface

Allows you to access specific information about DimXpert intersect points on the DimXpertManager tab of the Management Panel.  
**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 **.NET Syntax**

Visual Basic (Declaration)

Public Interface IDimXpertIntersectPointFeature

Visual Basic (Usage)  [Copy Code](#)

Dim instance As [IDimXpertIntersectPointFeature](#)

C#

public interface IDimXpertIntersectPointFeature

C++/CLI

public interface class IDimXpertIntersectPointFeature

 **Visual Basic for Applications (VBA) Syntax**

See [DimXpertIntersectPointFeature](#).

 **Example**

[Get DimXpert Intersect Features Example \(VBA\)](#)  
[Get DimXpert Intersect Features Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

 **Remarks**

You can access more general DimXpert feature information using [IDimXpertFeature](#).

 **Accessors**

[IDimXpertPart::GetFeatures](#) and [IDimXpertPart::!GetFeatures](#)

[IDimXpertPart::GetFeature](#)

[IFeature::GetSpecificFeature2](#)

 **See Also**

[IDimXpertIntersectPointFeature Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

**IDimXpertIntersectPointFeature Interface Members**  
[See Also](#) [Methods](#)

 

SolidWorks.Interop.swdimxpert Namespace : IDimXpertIntersectPointFeature Interface

The following tables list the members exposed by [IDimXpertIntersectPointFeature](#).

 **Public Methods**

| Name                                                                                                                 | Description                                                  |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
|  <a href="#">GetFeatures</a>      | Gets the DimXpert features of this DimXpert intersect point. |
|  <a href="#">GetNominalPoint</a>  | Gets the coordinates of this DimXpert intersect point.       |
|  <a href="#">GetNominalVector</a> | Gets the direction vector of this DimXpert intersect point.  |

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 See Also

[IDimXpertIntersectPointFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**IDimXpertIntersectPointFeature Interface Methods**  
[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertIntersectPointFeature Interface

For a list of all members of this type, see [IDimXpertIntersectPointFeature members](#).

 Public Methods

| Name                                                                                                               | Description                                                  |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
|  <a href="#">GetFeatures</a>      | Gets the DimXpert features of this DimXpert intersect point. |
|  <a href="#">GetNominalPoint</a>  | Gets the coordinates of this DimXpert intersect point.       |
|  <a href="#">GetNominalVector</a> | Gets the direction vector of this DimXpert intersect point.  |

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 See Also

[IDimXpertIntersectPointFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**GetFeatures Method (IDimXpertIntersectPointFeature)**  
[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertIntersectPointFeature Interface](#) : GetFeatures Method (IDimXpertIntersectPointFeature)

Gets the DimXpert features of this DimXpert intersect point.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetFeatures(  
    ByRef Feature1 As DimXpertFeature, _  
    ByRef Feature2 As DimXpertFeature _  
) As System.Boolean
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IDimXpertIntersectPointFeature  
Dim Feature1 As DimXpertFeature  
Dim Feature2 As DimXpertFeature  
Dim value As System.Boolean  
  
value = instance.GetFeatures(Feature1, Feature2)
```

C#

```
System.bool GetFeatures(  
    out DimXpertFeature Feature1,  
    out DimXpertFeature Feature2  
)
```

C++/CLI

```
System.bool GetFeatures(  
& [Out] DimXpertFeature^ Feature1,  
& [Out] DimXpertFeature^ Feature2  
)
```

**Parameters**

*Feature1*  
[IDimXpertFeature](#)  
*Feature2*  
[IDimXpertFeature](#)

## Return Value

True if method call is successful; false otherwise

## Visual Basic for Applications (VBA) Syntax

See [DimXpertIntersectPointFeature::GetFeatures](#).

## See Also

[IDimXpertIntersectPointFeature Interface](#)  
[IDimXpertIntersectPointFeature Members](#)

## Availability

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**GetNominalPoint Method (IDimXpertIntersectPointFeature)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertIntersectPointFeature Interface](#) : GetNominalPoint Method (IDimXpertIntersectPointFeature)

Gets the coordinates of this DimXpert intersect point.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetNominalPoint( _  
    ByRef X As System.Double, _  
    ByRef Y As System.Double, _  
    ByRef Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertIntersectPointFeature  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean  
  
value = instance.GetNominalPoint(X, Y, Z)
```

C#

```
System.bool GetNominalPoint(  
    out System.double X,  
    out System.double Y,  
    out System.double Z  
)
```

C++/CLI

```
System.bool GetNominalPoint(  
& [Out] System.double X,  
& [Out] System.double Y,  
& [Out] System.double Z  
)
```

## Parameters

[X](#)  
X-coordinate of the intersect point  
[Y](#)  
Y-coordinate of the intersect point  
[Z](#)  
Z-coordinate of the intersect point

## Return Value

True if method call is successful; false otherwise

## Visual Basic for Applications (VBA) Syntax

See [DimXpertIntersectPointFeature::GetNominalPoint](#).

## Example

[Get DimXpert Intersect Features Example \(VBA\)](#)  
[Get DimXpert Intersect Features Example \(VB.NET\)](#)

## See Also

[IDimXpertIntersectPointFeature Interface](#)  
[IDimXpertIntersectPointFeature Members](#)

## Availability

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**GetNominalVector Method (IDimXpertIntersectPointFeature)**

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertIntersectPointFeature Interface](#) : GetNominalVector Method (IDimXpertIntersectPointFeature)

Gets the direction vector of this DimXpert intersect point.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetNominalVector( _  
    ByRef I As System.Double, _  
    ByRef J As System.Double, _  
    ByRef K As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertIntersectPointFeature  
Dim I As System.Double  
Dim J As System.Double  
Dim K As System.Double  
Dim value As System.Boolean  
  
value = instance.GetNominalVector(I, J, K)
```

C#

```
System.bool GetNominalVector(  
    out System.double I,  
    out System.double J,  
    out System.double K  
)
```

C++/CLI

```
System.bool GetNominalVector(  
& [Out] System.double I,  
& [Out] System.double J,  
& [Out] System.double K  
)
```

### Parameters

*I*  
I component of the direction vector of the intersect point  
*J*  
J component of the direction vector of the intersect point  
*K*  
K component of the direction vector of the intersect point

### Return Value

True if method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertIntersectPointFeature::GetNominalVector](#).

#### See Also

[IDimXpertIntersectPointFeature Interface](#)  
[IDimXpertIntersectPointFeature Members](#)

#### Availability

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# IDimXpertLineProfileTolerance (Type)

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SolidWorks.Interop.swdimxpert Namespace : IDimXpertLineProfileTolerance Interface

Allows you to access DimXpert line profile tolerances.  
**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 .NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertLineProfileTolerance

Visual Basic (Usage)  Copy Code

Dim instance As [IDimXpertLineProfileTolerance](#)

C#

public interface IDimXpertLineProfileTolerance

C++/CLI

public interface class IDimXpertLineProfileTolerance

 Visual Basic for Applications (VBA) Syntax

See [DimXpertLineProfileTolerance](#).

 Example

[Get DimXpert Tolerance4 Example \(VBA\)](#)  
[Get DimXpert Tolerance4 Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

 Remarks

You can access more general geometric tolerance information using [IDimXpertTolerance](#).

 Accessors

[IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::!GetAppliedAnnotations](#)  
[IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::!GetAnnotations](#)  
[IDimXpertPart::GetAnnotation](#)  
[IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::!GetAppliedAnnotations](#)  
[IFeature::GetSpecificFeature2](#)

 See Also

[IDimXpertLineProfileTolerance Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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SolidWorks.Interop.swdimxpert Namespace : IDimXpertLineProfileTolerance Interface

The following tables list the members exposed by [IDimXpertLineProfileTolerance](#).

 Public Properties

| Name | Description |
|------|-------------|
|------|-------------|

|                                                                                                                      |                                                                          |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|  <a href="#">OuterToleranceValue</a> | Gets the outer tolerance value for this DimXpert line profile tolerance. |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|

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 **Public Methods**

| Name                                                                                                                  | Description                                                                               |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|  <a href="#">GetPlanarZoneVector</a> | Gets the planar zone vector that is used to compute this DimXpert line profile tolerance. |

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 **See Also**

[IDimXpertLineProfileTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**IDimXpertLineProfileTolerance Interface Methods**  
[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertLineProfileTolerance Interface

For a list of all members of this type, see [IDimXpertLineProfileTolerance members](#).

 **Public Methods**

| Name                                                                                                                  | Description                                                                               |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|  <a href="#">GetPlanarZoneVector</a> | Gets the planar zone vector that is used to compute this DimXpert line profile tolerance. |

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 **See Also**

[IDimXpertLineProfileTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**IDimXpertLineProfileTolerance Interface Properties**  
[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertLineProfileTolerance Interface

For a list of all members of this type, see [IDimXpertLineProfileTolerance members](#).

 **Public Properties**

| Name                                                                                                                    | Description                                                              |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|  <a href="#">OuterToleranceValue</a> | Gets the outer tolerance value for this DimXpert line profile tolerance. |

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 **See Also**

[IDimXpertLineProfileTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**GetPlanarZoneVector Method (IDimXpertLineProfileTolerance)**  
[See Also](#) [Example](#)

  [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertLineProfileTolerance Interface](#) : GetPlanarZoneVector Method (IDimXpertLineProfileTolerance)

Gets the planar zone vector that is used to compute this DimXpert line profile tolerance.

 **.NET Syntax**

Visual Basic (Declaration)

Function GetPlanarZoneVector( \_  
    ByRef [I](#) As System.Double, \_  
    ByRef [J](#) As System.Double, \_  
    ByRef [K](#) As System.Double \_  
) As System.Boolean

Visual Basic (Usage)

 [Copy Code](#)

```

Dim instance As IDimXpertLineProfileTolerance
Dim I As System.Double
Dim J As System.Double
Dim K As System.Double
Dim value As System.Boolean

value = instance.GetPlanarZoneVector(I, J, K)

```

C#

```

System.bool GetPlanarZoneVector(
    out System.double I,
    out System.double J,
    out System.double K
)

```

C++/CLI

```

System.bool GetPlanarZoneVector(
& [Out] System.double I,
& [Out] System.double J,
& [Out] System.double K
)

```

#### Parameters

*I*  
I component of the planar zone vector  
*J*  
J component of the planar zone vector  
*K*  
K component of the planar zone vector

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertLineProfileTolerance::GetPlanarZoneVector](#).

#### Example

[Get DimXpert Tolerance4 Example \(VBA\)](#)  
[Get DimXpert Tolerance4 Example \(VB.NET\)](#)

#### See Also

[IDimXpertLineProfileTolerance Interface](#)  
[IDimXpertLineProfileTolerance Members](#)

#### Availability

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**OuterToleranceValue Property (IDimXpertLineProfileTolerance)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertLineProfileTolerance Interface](#) : OuterToleranceValue Property (IDimXpertLineProfileTolerance)

Gets the outer tolerance value for this DimXpert line profile tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```

ReadOnly Property OuterToleranceValue As System.Double

```

Visual Basic (Usage)



```
Dim instance As IDimXpertLineProfileTolerance  
Dim value As System.Double  
  
value = instance.OuterToleranceValue
```

C#

```
System.double OuterToleranceValue {get;}
```

C++/CLI

```
property System.double OuterToleranceValue {  
    System.double get();  
}
```

#### Property Value

Outer tolerance value

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertLineProfileTolerance::OuterToleranceValue](#).

#### Example

[Get DimXpert Tolerance4 Example \(VBA\)](#)  
[Get DimXpert Tolerance4 Example \(VB.NET\)](#)

#### See Also

[IDimXpertLineProfileTolerance Interface](#)  
[IDimXpertLineProfileTolerance Members](#)

#### Availability

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# IDimXpertOrientationTolerance (Type)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertOrientationTolerance Interface

Allows you to access DimXpert orientation tolerances.  
**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 **NET Syntax**

Visual Basic (Declaration)

Public Interface IDimXpertOrientationTolerance

Visual Basic (Usage) 

Dim instance As [IDimXpertOrientationTolerance](#)

C#

public interface IDimXpertOrientationTolerance

C++/CLI

public interface class IDimXpertOrientationTolerance

 **Visual Basic for Applications (VBA) Syntax**

See [DimXpertOrientationTolerance](#).

 **Example**

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

 **Remarks**

This interface defines angularity, parallelism and perpendicularity geometric tolerances. You can access more general geometric tolerance information using [IDimXpertTolerance](#).

 **Accessors**

[IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::!GetAppliedAnnotations](#)  
[IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::!GetAnnotations](#)  
[IDimXpertPart::GetAnnotation](#)  
[IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::!GetAppliedAnnotations](#)  
[IFeature::GetSpecificFeature2](#)

 **See Also**

[IDimXpertOrientationTolerance Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertOrientationTolerance Interface

The following tables list the members exposed by [IDimXpertOrientationTolerance](#).

 **Public Properties**

| Name                                                                                                         | Description                                                                   |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|  <a href="#">Modifier</a> | Gets the material condition modifier for this DimXpert orientation tolerance. |

|                                                                                                           |                                                            |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|  <a href="#">ZoneType</a> | Gets the zone type of this DimXpert orientation tolerance. |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------|

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## Public Methods

| Name                                                                                                                  | Description                                                                              |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|  <a href="#">GetMaxTolerance</a>     | Gets the maximum tolerance for this DimXpert orientation tolerance.                      |
|  <a href="#">GetPlanarZoneVector</a> | Gets the planar zone vector that is used to compute this DimXpert orientation tolerance. |
|  <a href="#">GetProjectedZone</a>    | Gets the projected zone value of this DimXpert orientation tolerance.                    |
|  <a href="#">IsTangentPlane</a>      | Gets whether the tangent plane is specified for this DimXpert orientation tolerance.     |

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## See Also

[IDimXpertOrientationTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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### IDimXpertOrientationTolerance Interface Methods

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertOrientationTolerance Interface

For a list of all members of this type, see [IDimXpertOrientationTolerance members](#).

## Public Methods

| Name                                                                                                                  | Description                                                                              |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|  <a href="#">GetMaxTolerance</a>     | Gets the maximum tolerance for this DimXpert orientation tolerance.                      |
|  <a href="#">GetPlanarZoneVector</a> | Gets the planar zone vector that is used to compute this DimXpert orientation tolerance. |
|  <a href="#">GetProjectedZone</a>    | Gets the projected zone value of this DimXpert orientation tolerance.                    |
|  <a href="#">IsTangentPlane</a>     | Gets whether the tangent plane is specified for this DimXpert orientation tolerance.     |

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## See Also

[IDimXpertOrientationTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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### IDimXpertOrientationTolerance Interface Properties

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertOrientationTolerance Interface

For a list of all members of this type, see [IDimXpertOrientationTolerance members](#).

## Public Properties

| Name                                                                                                         | Description                                                                   |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|  <a href="#">Modifier</a> | Gets the material condition modifier for this DimXpert orientation tolerance. |
|  <a href="#">ZoneType</a> | Gets the zone type of this DimXpert orientation tolerance.                    |

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## See Also

[IDimXpertOrientationTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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### GetMaxTolerance Method (IDimXpertOrientationTolerance)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertOrientationTolerance Interface](#) : GetMaxTolerance Method (IDimXpertOrientationTolerance)

Gets the maximum tolerance for this DimXpert orientation tolerance.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetMaxTolerance( _
    ByRef IsMax As System.Boolean, _
    ByRef Tolerance As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertOrientationTolerance
Dim IsMax As System.Boolean
Dim Tolerance As System.Double
Dim value As System.Boolean

value = instance.GetMaxTolerance(IsMax, Tolerance)
```

C#

```
System.bool GetMaxTolerance(
    out System.bool IsMax,
    out System.double Tolerance
)
```

C++/CLI

```
System.bool GetMaxTolerance(
    & [Out] System.bool IsMax,
    & [Out] System.double Tolerance
)
```

#### Parameters

*IsMax*  
True if maximum tolerance is set; false otherwise  
*Tolerance*  
Maximum tolerance value

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertOrientationTolerance::GetMaxTolerance](#).

#### See Also

[IDimXpertOrientationTolerance Interface](#)  
[IDimXpertOrientationTolerance Members](#)

#### Availability

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#### GetPlanarZoneVector Method (IDimXpertOrientationTolerance)

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertOrientationTolerance Interface](#) : GetPlanarZoneVector Method (IDimXpertOrientationTolerance)

Gets the planar zone vector that is used to compute this DimXpert orientation tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetPlanarZoneVector( _
    ByRef I As System.Double, _
    ByRef J As System.Double, _
    ByRef K As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```

Dim instance As IDimXpertOrientationTolerance
Dim I As System.Double
Dim J As System.Double
Dim K As System.Double
Dim value As System.Boolean

value = instance.GetPlanarZoneVector(I, J, K)

```

C#

```

System.bool GetPlanarZoneVector(
    out System.double I,
    out System.double J,
    out System.double K
)

```

C++/CLI

```

System.bool GetPlanarZoneVector(
    & [Out] System.double I,
    & [Out] System.double J,
    & [Out] System.double K
)

```

#### Parameters

*I*  
 I component of the planar zone vector  
*J*  
 J component of the planar zone vector  
*K*  
 K component of the planar zone vector

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertOrientationTolerance::GetPlanarZoneVector](#).

#### Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

#### See Also

[IDimXpertOrientationTolerance Interface](#)  
[IDimXpertOrientationTolerance Members](#)

#### Availability

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**GetProjectedZone Method (IDimXpertOrientationTolerance)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertOrientationTolerance Interface](#) : GetProjectedZone Method (IDimXpertOrientationTolerance)

Gets the projected zone value of this DimXpert orientation tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```

Function GetProjectedZone( _
    ByRef Enabled As System.Boolean, _
    ByRef Value As System.Double _
) As System.Boolean

```

```
Dim instance As IDimXpertOrientationTolerance
Dim Enabled As System.Boolean
Dim Value As System.Double
Dim value As System.Boolean

value = instance.GetProjectedZone(Enabled, Value)
```

## C#

```
System.bool GetProjectedZone(
    out System.bool Enabled,
    out System.double Value
)
```

## C++/CLI

```
System.bool GetProjectedZone(
    & [Out] System.bool Enabled,
    & [Out] System.double Value
)
```

## Parameters

*Enabled*

True if enabled; false otherwise

*Value*

Projected zone value

## Return Value

True if the method call is successful; false otherwise

 Visual Basic for Applications (VBA) Syntax

See [DimXpertOrientationTolerance::GetProjectedZone](#).

 Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

 See Also

[IDimXpertOrientationTolerance Interface](#)  
[IDimXpertOrientationTolerance Members](#)

 Availability

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SOLIDWORKS DimXpert API Help

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**IsTangentPlane Method (IDimXpertOrientationTolerance)**

See Also [Example](#)



SolidWorks.Interop.swdimxpert Namespace > [IDimXpertOrientationTolerance Interface](#) : IsTangentPlane Method (IDimXpertOrientationTolerance)

Gets whether the tangent plane is specified for this DimXpert orientation tolerance.

 .NET Syntax

## Visual Basic (Declaration)

```
Function IsTangentPlane() As System.Boolean
```

## Visual Basic (Usage)

```
Dim instance As IDimXpertOrientationTolerance
Dim value As System.Boolean
```

```
value = instance.IsTangentPlane()
```

C#

```
System.bool IsTangentPlane()
```

C++/CLI

```
System.bool IsTangentPlane();
```

#### Return Value

True if tangent plane is specified; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertOrientationTolerance::IsTangentPlane](#).

#### Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

#### See Also

[IDimXpertOrientationTolerance Interface](#)  
[IDimXpertOrientationTolerance Members](#)

#### Availability

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**Modifier Property (IDimXpertOrientationTolerance)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertOrientationTolerance Interface](#) : Modifier Property (IDimXpertOrientationTolerance)

Gets the material condition modifier for this DimXpert orientation tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Modifier As swDimXpertMaterialConditionModifier\_e
```

Visual Basic (Usage)



```
Dim instance As IDimXpertOrientationTolerance  
Dim value As swDimXpertMaterialConditionModifier\_e  
  
value = instance.Modifier
```

C#

```
swDimXpertMaterialConditionModifier\_e Modifier {get;}
```

C++/CLI

```
property swDimXpertMaterialConditionModifier\_e Modifier {  
    swDimXpertMaterialConditionModifier\_e get();  
}
```

#### Property Value

Material condition modifier as defined in [swDimXpertMaterialConditionModifier\\_e](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertOrientationTolerance::Modifier](#).

#### Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

#### See Also

[IDimXpertOrientationTolerance Interface](#)  
[IDimXpertOrientationTolerance Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

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**ZoneType Property (IDimXpertOrientationTolerance)**  
[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertOrientationTolerance Interface](#) : ZoneType Property (IDimXpertOrientationTolerance)

Gets the zone type of this DimXpert orientation tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ZoneType As swDimXpert0rientationZoneType\_e
```

Visual Basic (Usage)



```
Dim instance As IDimXpert0rientationTolerance  
Dim value As swDimXpert0rientationZoneType\_e  
  
value = instance.ZoneType
```

C#

```
swDimXpert0rientationZoneType\_e ZoneType {get;}
```

C++/CLI

```
property swDimXpert0rientationZoneType\_e ZoneType {  
    swDimXpert0rientationZoneType\_e get();  
}
```

#### Property Value

Orientation zone type as defined in [swDimXpertOrientationZoneType\\_e](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertOrientationTolerance::ZoneType](#).

#### Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

#### See Also

[IDimXpertOrientationTolerance Interface](#)  
[IDimXpertOrientationTolerance Members](#)

#### Availability

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IDimXpertPart (Type)

SOLIDWORKS DimXpert API Help

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IDimXpertPart Interface

[See Also](#) [Members](#) [Example](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPart Interface

Allows you to access the geometric dimensioning and tolerancing information that underpins the DimXpert functionality, features, and annotations for a given model.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

Visual Basic Syntax

Visual Basic (Declaration)

Public Interface IDimXpertPart

Visual Basic (Usage)



Dim instance As [IDimXpertPart](#)

C#

public interface IDimXpertPart

C++/CLI

public interface class IDimXpertPart

Visual Basic for Applications (VBA) Syntax

See [DimXpertPart](#).

Example

- [Get and Set Pattern \(C#\)](#)
- [Get and Set Pattern \(VB.NET\)](#)
- [Get and Set Pattern \(VBA\)](#)
- [Get and Set Size Dimension \(C#\)](#)
- [Get and Set Size Dimension \(VB.NET\)](#)
- [Get and Set Size Dimension \(VBA\)](#)
- [Get and Set Location Dimension \(C#\)](#)
- [Get and Set Location Dimension \(VB.NET\)](#)
- [Get and Set Location Dimension \(VBA\)](#)
- [Get and Set Datum \(C#\)](#)
- [Get and Set Datum \(VB.NET\)](#)
- [Get and Set Datum \(VBA\)](#)
- [Auto Dimension Scheme \(C#\)](#)
- [Auto Dimension Scheme \(VB.NET\)](#)
- [Auto Dimension Scheme \(VBA\)](#)
- [Get DimXpert Feature \(VBA\)](#)
- [Get DimXpert Feature \(VB.NET\)](#)
- [Get DimXpert Features and Annotations \(C#\)](#)

Accessors

[IDimXpertManager::DimXpertPart property](#)

See Also

- [IDimXpertPart Members](#)
- [SolidWorks.Interop.swdimxpert Namespace](#)

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IDimXpertPart Interface Methods

[See Also](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPart Interface

For a list of all members of this type, see [IDimXpertPart members](#).

Public Methods

| Name                                | Description                            |
|-------------------------------------|----------------------------------------|
| <a href="#">AutoDimensionScheme</a> | Creates an Auto Dimension Scheme.      |
| <a href="#">DeleteAllTolerances</a> | Removes all tolerances from the model. |

|                                                                                   |                                         |                                                                                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
|  |                                         |                                                                                                       |
|  | <a href="#">GetAnnotation</a>           | Gets a DimXpert annotation by name for the given model.                                               |
|  | <a href="#">GetAnnotationCount</a>      | Gets the total number of DimXpert annotations in the model.                                           |
|  | <a href="#">GetAnnotations</a>          | Gets all of the DimXpert annotations in the model.                                                    |
|  | <a href="#">GetAutoDimSchemeOption</a>  | Gets options for creation of an Auto Dimension Scheme.                                                |
|  | <a href="#">GetBlockTolerances</a>      | Gets all of the DimXpert block tolerances in the model.                                               |
|  | <a href="#">GetDimOption</a>            | Gets options for insertion of datums, patterns, and dimensions.                                       |
|  | <a href="#">GetFeature</a>              | Gets a DimXpert feature by name for the given model.                                                  |
|  | <a href="#">GetFeatureCount</a>         | Gets the total number of DimXpert features in the model.                                              |
|  | <a href="#">GetFeatures</a>             | Gets all of the DimXpert features in the model.                                                       |
|  | <a href="#">IGetAnnotations</a>         | Gets all of the DimXpert annotations in the model.                                                    |
|  | <a href="#">IGetFeatures</a>            | Gets all of the DimXpert features in the model.                                                       |
|  | <a href="#">ImportScheme</a>            | Imports the tolerance scheme of the specified file to the specified configuration of this open model. |
|  | <a href="#">InsertAngleDimension</a>    | Inserts an angle dimension.                                                                           |
|  | <a href="#">InsertBasicDimension</a>    | Inserts a basic dimension.                                                                            |
|  | <a href="#">InsertDatum</a>             | Inserts a datum for the selected face.                                                                |
|  | <a href="#">InsertGtol</a>              | Inserts the specified Gtol annotation.                                                                |
|  | <a href="#">InsertLocationDimension</a> | Inserts a linear or angular location dimension between a selected feature and its locating feature.   |
|  | <a href="#">InsertPattern</a>           | Inserts a linked, manual, or collection pattern using the selected faces.                             |
|  | <a href="#">InsertSizeDimension</a>     | Inserts a size dimension for the selected face or edge.                                               |

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## See Also

[IDimXpertPart Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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## IDimXpertPart Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPart Interface

The following tables list the members exposed by [IDimXpertPart](#).

## Public Properties

| Name                                                                                                                    | Description                                                       |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|  <a href="#">ShowToleranceStatus</a> | Gets and sets whether tolerance status is displayed in the model. |

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## Public Methods

| Name                                                                                                                       | Description                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|  <a href="#">AutoDimensionScheme</a>    | Creates an Auto Dimension Scheme.                                                                     |
|  <a href="#">DeleteAllTolerances</a>    | Removes all tolerances from the model.                                                                |
|  <a href="#">GetAnnotation</a>          | Gets a DimXpert annotation by name for the given model.                                               |
|  <a href="#">GetAnnotationCount</a>     | Gets the total number of DimXpert annotations in the model.                                           |
|  <a href="#">GetAnnotations</a>         | Gets all of the DimXpert annotations in the model.                                                    |
|  <a href="#">GetAutoDimSchemeOption</a> | Gets options for creation of an Auto Dimension Scheme.                                                |
|  <a href="#">GetBlockTolerances</a>     | Gets all of the DimXpert block tolerances in the model.                                               |
|  <a href="#">GetDimOption</a>           | Gets options for insertion of datums, patterns, and dimensions.                                       |
|  <a href="#">GetFeature</a>             | Gets a DimXpert feature by name for the given model.                                                  |
|  <a href="#">GetFeatureCount</a>        | Gets the total number of DimXpert features in the model.                                              |
|  <a href="#">GetFeatures</a>            | Gets all of the DimXpert features in the model.                                                       |
|  <a href="#">IGetAnnotations</a>        | Gets all of the DimXpert annotations in the model.                                                    |
|  <a href="#">IGetFeatures</a>           | Gets all of the DimXpert features in the model.                                                       |
|  <a href="#">ImportScheme</a>           | Imports the tolerance scheme of the specified file to the specified configuration of this open model. |
|  <a href="#">InsertAngleDimension</a>   | Inserts an angle dimension.                                                                           |

|                                                                                   |                                         |                                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
|   |                                         |                                                                                                     |
|  | <a href="#">InsertBasicDimension</a>    | Inserts a basic dimension.                                                                          |
|  | <a href="#">InsertDatum</a>             | Inserts a datum for the selected face.                                                              |
|  | <a href="#">InsertGtol</a>              | Inserts the specified Gtol annotation.                                                              |
|  | <a href="#">InsertLocationDimension</a> | Inserts a linear or angular location dimension between a selected feature and its locating feature. |
|  | <a href="#">InsertPattern</a>           | Inserts a linked, manual, or collection pattern using the selected faces.                           |
|  | <a href="#">InsertSizeDimension</a>     | Inserts a size dimension for the selected face or edge.                                             |

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 See Also

[IDimXpertPart Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**IDimXpertPart Interface Properties**

[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : [IDimXpertPart Interface](#)

For a list of all members of this type, see [IDimXpertPart members](#).

 Public Properties

| Name                                                                                                                  | Description                                                       |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|  <a href="#">ShowToleranceStatus</a> | Gets and sets whether tolerance status is displayed in the model. |

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 See Also

[IDimXpertPart Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**AutoDimensionScheme Method (IDimXpertPart)**

[See Also](#) [Example](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : AutoDimensionScheme Method (IDimXpertPart)  
Creates an Auto Dimension Scheme.

 .NET Syntax

Visual Basic (Declaration)

```
Function AutoDimensionScheme( _
    ByVal Option As DimXpertAutoDimSchemeOption _
) As System.Boolean
```

Visual Basic (Usage)  Copy Code

```
Dim instance As IDimXpertPart
Dim Option As DimXpertAutoDimSchemeOption
Dim value As System.Boolean

value = instance.AutoDimensionScheme(Option)
```

C#

```
System.bool AutoDimensionScheme(
    DimXpertAutoDimSchemeOption Option
)
```

C++/CLI

```
System.bool AutoDimensionScheme(
    & DimXpertAutoDimSchemeOption^ Option
)
```

**Parameters**

*Option*  
[IDimXpertAutoDimSchemeOption](#)

#### Return Value

True if Auto Dimension Scheme is created, false if not

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::AutoDimensionScheme](#).

#### Example

[Auto Dimension Scheme \(C#\)](#)  
[Auto Dimension Scheme \(VB.NET\)](#)  
[Auto Dimension Scheme \(VBA\)](#)

#### Remarks

Before calling this method, call [IDimXpertPart::GetAutoDimSchemeOption](#) to populate the Option parameter.

#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart::InsertBasicDimension Method \(\)](#)  
[IDimXpertPart::InsertDatum Method \(\)](#)  
[IDimXpertPart::InsertLocationDimension Method \(\)](#)  
[IDimXpertPart::InsertPattern Method \(\)](#)  
[IDimXpertPart::InsertSizeDimension Method \(\)](#)

#### Availability

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#### DeleteAllTolerances Method (IDimXpertPart)

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : DeleteAllTolerances Method (IDimXpertPart)

Removes all tolerances from the model.

#### .NET Syntax

##### Visual Basic (Declaration)

```
Function DeleteAllTolerances() As System.Boolean
```

##### Visual Basic (Usage)



```
Dim instance As IDimXpertPart
Dim value As System.Boolean

value = instance.DeleteAllTolerances()
```

##### C#

```
System.bool DeleteAllTolerances()
```

##### C++/CLI

```
System.bool DeleteAllTolerances();
```

#### Return Value

True if all tolerances were removed; false if not

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::DeleteAllTolerances](#).

## Example

[Auto Dimension Scheme \(C#\)](#)  
[Auto Dimension Scheme \(VB.NET\)](#)  
[Auto Dimension Scheme \(VBA\)](#)

## See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)

## Availability

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**GetAnnotation Method (IDimXpertPart)**

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetAnnotation Method (IDimXpertPart)

Gets a DimXpert annotation by name for the given model.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetAnnotation( _  
    ByVal Name As System.String _  
) As DimXpertAnnotation
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Name As System.String  
Dim value As DimXpertAnnotation  
  
value = instance.GetAnnotation(Name)
```

### C#

```
DimXpertAnnotation GetAnnotation(  
    System.string Name  
)
```

### C++/CLI

```
DimXpertAnnotation^ GetAnnotation(  
& System.String^ Name  
)
```

### Parameters

*Name*  
Name of the annotation

### Return Value

[IDimXpertAnnotation](#)

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::GetAnnotation](#).

## See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)

## Availability

**GetAnnotationCount Method (IDimXpertPart)**[See Also](#)[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetAnnotationCount Method (IDimXpertPart)

Gets the total number of DimXpert annotations in the model.

**.NET Syntax****Visual Basic (Declaration)**

```
Function GetAnnotationCount() As System.Integer
```

**Visual Basic (Usage)**

```
Dim instance As IDimXpertPart
Dim value As System.Integer

value = instance.GetAnnotationCount()
```

**C#**

```
System.int GetAnnotationCount()
```

**C++/CLI**

```
System.int GetAnnotationCount();
```

**Return Value**

Number of DimXpert annotations

**Visual Basic for Applications (VBA) Syntax**See [DimXpertPart::GetAnnotationCount](#). **See Also**[IDimXpertPart Interface](#)[IDimXpertPart Members](#) **Availability****GetAnnotations Method (IDimXpertPart)**[See Also](#)[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetAnnotations Method (IDimXpertPart)

Gets all of the DimXpert annotations in the model.

**.NET Syntax****Visual Basic (Declaration)**

```
Function GetAnnotations() As System.Object
```

**Visual Basic (Usage)**

```
Dim instance As IDimXpertPart
Dim value As System.Object

value = instance.GetAnnotations()
```

**C#**

```
System.Object GetAnnotations()
```

C++/CLI

```
System.Object^ GetAnnotations();
```

#### Return Value

Array of [IDimXpertAnnotations](#).

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::GetAnnotations](#).

#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)

#### Availability

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#### GetAutoDimSchemeOption Method (IDimXpertPart)

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetAutoDimSchemeOption Method (IDimXpertPart)

Gets options for creation of an Auto Dimension Scheme.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetAutoDimSchemeOption() As DimXpertAutoDimSchemeOption
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim value As DimXpertAutoDimSchemeOption  
  
value = instance.GetAutoDimSchemeOption()
```

C#

```
DimXpertAutoDimSchemeOption GetAutoDimSchemeOption()
```

C++/CLI

```
DimXpertAutoDimSchemeOption^ GetAutoDimSchemeOption();
```

#### Return Value

[IDimXpertAutoDimSchemeOption](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::GetAutoDimSchemeOption](#).

#### Example

[Auto Dimension Scheme \(C#\)](#)  
[Auto Dimension Scheme \(VB.NET\)](#)  
[Auto Dimension Scheme \(VBA\)](#)

#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart::AutoDimensionScheme Method](#)

## Availability

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### GetBlockTolerances Method (IDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetBlockTolerances Method (IDimXpertPart)

Gets all of the DimXpert block tolerances in the model.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetBlockTolerances() As DimXpertBlockTolerances
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim value As DimXpertBlockTolerances  
  
value = instance.GetBlockTolerances()
```

C#

```
DimXpertBlockTolerances GetBlockTolerances()
```

C++/CLI

```
DimXpertBlockTolerances^ GetBlockTolerances();
```

### Return Value

[IDimXpertBlockTolerances](#)

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::GetBlockTolerances](#).

## See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)

## Availability

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### GetDimOption Method (IDimXpertPart)

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetDimOption Method (IDimXpertPart)

Gets options for insertion of datums, patterns, and dimensions.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetDimOption() As DimXpertDimensionOption
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart
Dim value As DimXpertDimensionOption

value = instance.GetDimOption()
```

C#

```
DimXpertDimensionOption GetDimOption()
```

C++/CLI

```
DimXpertDimensionOption^ GetDimOption();
```

#### Return Value

[IDimXpertDimensionOption](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::GetDimOption](#).

#### Example

[Get and Set Pattern \(C#\)](#)  
[Get and Set Pattern \(VB.NET\)](#)  
[Get and Set Pattern \(VBA\)](#)

#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart::InsertDatum Method](#)  
[IDimXpertPart::InsertLocationDimension Method](#)  
[IDimXpertPart::InsertPattern Method](#)  
[IDimXpertPart::InsertSizeDimension Method](#)  
[IDimXpertPart::InsertBasicDimension Method](#)

#### Availability

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#### GetFeature Method (IDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetFeature Method (IDimXpertPart)

Gets a DimXpert feature by name for the given model.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetFeature( _
    ByVal Name As System.String _
) As DimXpertFeature
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart
Dim Name As System.String
Dim value As DimXpertFeature

value = instance.GetFeature(Name)
```

C#

```
DimXpertFeature GetFeature(
    System.string Name
)
```

C++/CLI

```
DimXpertFeature^ GetFeature(  
& System.String^ Name  
)
```

#### Parameters

*Name*

Name of a DimXpert feature

#### Return Value

[IDimXpertFeature](#)



#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::GetFeature](#).



#### See Also

[IDimXpertPart Interface](#)

[IDimXpertPart Members](#)



#### Availability

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#### GetFeatureCount Method (IDimXpertPart)

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetFeatureCount Method (IDimXpertPart)

Gets the total number of DimXpert features in the model.



#### .NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart
```

```
Dim value As System.Integer
```

```
value = instance.GetFeatureCount()
```

C#

```
System.int GetFeatureCount()
```

C++/CLI

```
System.int GetFeatureCount();
```

#### Return Value

Number of DimXpert features



#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::GetFeatureCount](#)



#### Example

[Get DimXpert Feature \(VBA\)](#)

[Get DimXpert Feature \(VB.NET\)](#)

## See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)

## Availability

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### GetFeatures Method (IDimXpertPart)

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : GetFeatures Method (IDimXpertPart)

Gets all of the DimXpert features in the model.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetFeatures() As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim value As System.Object  
  
value = instance.GetFeatures()
```

### C#

```
System.Object GetFeatures()
```

### C++/CLI

```
System.Object^ GetFeatures();
```

### Return Value

Array of [IDimXpertFeatures](#)

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::GetFeatures](#).

## Example

[Get DimXpert Feature \(VBA\)](#)  
[Get DimXpert Feature \(VB.NET\)](#)

## See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)

## Availability

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### IGetAnnotations Method (IDimXpertPart)

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : IGetAnnotations Method (IDimXpertPart)

Gets all of the DimXpert annotations in the model.

## .NET Syntax

### Visual Basic (Declaration)

```
Function IGetAnnotations( _  
    ByVal Count As System.Integer _  
) As DimXpertAnnotation
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Count As System.Integer  
Dim value As DimXpertAnnotation  
  
value = instance.IGetAnnotations(Count)
```

C#

```
DimXpertAnnotation IGetAnnotations(  
    System.int Count  
)
```

C++/CLI

```
DimXpertAnnotation^ IGetAnnotations(  
&    System.int Count  
)
```

#### Parameters

*Count*  
Number of DimXpert annotations

#### Return Value

- in-process, unmanaged C++; Pointer to an array of [IDimXpertAnnotation](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

#### Remarks

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertPart::GetAnnotationCount](#) to get the value for the Count parameter.

#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)

#### Availability

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**IGetFeatures Method (IDimXpertPart)**

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : IGetFeatures Method (IDimXpertPart)

Gets all of the DimXpert features in the model.

#### .NET Syntax

Visual Basic (Declaration)

```
Function IGetFeatures( _  
    ByVal Count As System.Integer _  
) As DimXpertFeature
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Count As System.Integer  
Dim value As DimXpertFeature
```

```
value = instance.IGetFeatures(Count)
```

C#

```
DimXpertFeature IGetFeatures(  
    System.int Count  
)
```

C++/CLI

```
DimXpertFeature^ IGetFeatures(  
& System.int Count  
)
```

#### Parameters

*Count*

Number of DimXpert features

#### Return Value

- in-process, unmanaged C++: Pointer to an array of [IDimXpertFeature](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

#### Remarks

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertPart::GetFeatureCount](#) to get the value for the Count parameter.

#### See Also

[IDimXpertPart Interface](#)

[IDimXpertPart Members](#)

#### Availability

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**ImportScheme Method (IDimXpertPart)**

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : ImportScheme Method (IDimXpertPart)

Imports the tolerance scheme of the specified file to the specified configuration of this open model.

#### NET Syntax

Visual Basic (Declaration)

```
Function ImportScheme( _  
    ByVal Filename As System.String, _  
    ByVal ConfigurationName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Filename As System.String  
Dim ConfigurationName As System.String  
Dim value As System.Boolean  
  
value = instance.ImportScheme(Filename, ConfigurationName)
```

C#

```
System.bool ImportScheme(
    System.string Filename,
    System.string ConfigurationName
)
```

C++/CLI

```
System.bool ImportScheme(
    & System.String^ Filename,
    & System.String^ ConfigurationName
)
```

#### Parameters

*Filename*

Full path name of the file whose tolerance scheme to import to ConfigurationName

*ConfigurationName*

Name of the configuration in the open model document to which to import the tolerance scheme

#### Return Value

True if tolerance scheme successfully imported, false if not

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart.ImportScheme](#).

#### Example

'VBA

'1. Open a model document with a Default configuration.

'2. Edit the ImportScheme command to point to a model document with the tolerance scheme to import.

'3. Run the macro.

Option Explicit

Dim swApp As SldWorks.SldWorks

Dim swModel As SldWorks.ModelDoc2

Dim swModelDocExt As SldWorks.ModelDocExtension

Dim swSchema As SldWorks.DimXpertManager

Dim swDXPart As DimXpertPart

Dim status As Boolean

Sub main()

Set swApp = Application.SldWorks

Set swModel = swApp.ActiveDoc

Set swModelDocExt = swModel.Extension

Set swSchema = swModelDocExt.DimXpertManager("Default", True)

Set swDXPart = swSchema.DimXpertPart

status = swDXPart.**ImportScheme**("Path\_name\model\_name.SLDPRT", "Default")

End Sub

#### See Also

[IDimXpertPart Interface](#)

[IDimXpertPart Members](#)

#### Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

**InsertAngleDimension Method (IDimXpertPart)**

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : InsertAngleDimension Method (IDimXpertPart)

Inserts an angle dimension.

## .NET Syntax

### Visual Basic (Declaration)

```
Function InsertAngleDimension( _  
    ByVal Option As DimXpertDimensionOption _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Option As DimXpertDimensionOption  
Dim value As System.Boolean  
  
value = instance.InsertAngleDimension(Option)
```

### C#

```
System.bool InsertAngleDimension(  
    DimXpertDimensionOption Option  
)
```

### C++/CLI

```
System.bool InsertAngleDimension(  
& DimXpertDimensionOption^ Option  
)
```

#### Parameters

*Option*  
[IDimXpertDimensionOption](#)

#### Return Value

True if the angle dimension is created, false if not

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPart.InsertAngleDimension](#).

## See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart.InsertBasicDimension Method \(\)](#)

## Availability

SOLIDWORKS 2026 SP02, Revision Number 34.2

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

**InsertBasicDimension Method (IDimXpertPart)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : InsertBasicDimension Method (IDimXpertPart)

Inserts a basic dimension.

## .NET Syntax

### Visual Basic (Declaration)

```
Function InsertBasicDimension( _  
    ByVal Option As DimXpertDimensionOption _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Option As DimXpertDimensionOption
```

```
Dim value As System.Boolean  
value = instance.InsertBasicDimension(Option)
```

C#

```
System.bool InsertBasicDimension(  
    DimXpertDimensionOption Option  
)
```

C++/CLI

```
System.bool InsertBasicDimension(  
& DimXpertDimensionOption^ Option  
)
```

#### Parameters

*Option*  
[IDimXpertDimensionOption](#)

#### Return Value

True if the basic dimension is created, false if not

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::InsertBasicDimension](#).

#### Example

[Insert Basic Dimension \(VBA\)](#)  
[Insert Basic Dimension \(VB.NET\)](#)  
[Insert Basic Dimension \(C#\)](#)

#### Remarks

A DimXpert basic dimension is the distance or angle between selected reference features.

Before calling this method:

1. Call [IDimXpertPart::GetDimOption](#) to get an instance of IDimXpertDimensionOption.
2. Set:
  - IDimXpertDimensionOption::TextPosition or [IDimXpertDimensionOption::DimensionPositionOption](#) (if both are set, only DimensionPositionOption is in force) for distance-between dimensions.
  - [IDimXpertDimensionOption::TextPosition](#) for angle-between dimensions.
3. Populate Option with the instance of IDimXpertDimensionOption.
4. Multi-select reference features using [IModelDocExtension::SelectByID2](#):
  - a. Origin feature (Append = false)
  - b. Tolerance feature (Append = true)

#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart::InsertDatum Method \(\)](#)  
[IDimXpertPart::InsertLocationDimension Method \(\)](#)  
[IDimXpertPart::InsertPattern Method \(\)](#)  
[IDimXpertPart::InsertSizeDimension Method \(\)](#)  
[IDimXpertPart::AutoDimensionScheme Method \(\)](#)  
[IDimXpertPart::InsertAngleDimension Method \(\)](#)

#### Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

#### InsertDatum Method (IDimXpertPart)

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : InsertDatum Method (IDimXpertPart)

Inserts a datum for the selected face.

#### .NET Syntax

Visual Basic (Declaration)

```
Function InsertDatum( _  
    ByVal Option As DimXpertDimensionOption _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Option As DimXpertDimensionOption  
Dim value As System.Boolean  
  
value = instance.InsertDatum(Option)
```

C#

```
System.bool InsertDatum(  
    DimXpertDimensionOption Option  
)
```

C++/CLI

```
System.bool InsertDatum(  
& DimXpertDimensionOption^ Option  
)
```

#### Parameters

*Option*

[IDimXpertDimensionOption](#)

#### Return Value

True if datum is created, false if not



#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart.InsertDatum](#).



#### Example

[Get and Set Datum \(C#\)](#)  
[Get and Set Datum \(VB.NET\)](#)  
[Get and Set Datum \(VBA\)](#)



#### Remarks

Before calling this method:

1. Call [IDimXpertPart.GetDimOption](#) to get an instance of IDimXpertDimensionOption.
2. Set [IDimXpertDimensionOption.FeatureSelectorOptions](#).
3. Populate Option with the instance of IDimXpertDimensionOption.
4. Select a face to which to attach the datum.



#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart.AutoDimensionScheme Method \(\)](#)  
[IDimXpertPart.InsertBasicDimension Method \(\)](#)  
[IDimXpertPart.InsertLocationDimension Method \(\)](#)  
[IDimXpertPart.InsertPattern Method \(\)](#)  
[IDimXpertPart.InsertSizeDimension Method \(\)](#)



#### Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

**InsertGtol Method (IDimXpertPart)**

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : InsertGtol Method (IDimXpertPart)

Inserts the specified Gtol annotation.

## .NET Syntax

### Visual Basic (Declaration)

```
Function InsertGtol( _  
    ByVal Type As System.Integer _  
) As DimXpertAnnotation
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Type As System.Integer  
Dim value As DimXpertAnnotation  
  
value = instance.InsertGtol(Type)
```

### C#

```
DimXpertAnnotation InsertGtol(  
    System.int Type  
)
```

### C++/CLI

```
DimXpertAnnotation^ InsertGtol(  
& System.int Type  
)
```

#### Parameters

Type

Type of Gtol to insert as defined in [swDimXpertGtolType\\_e](#)

#### Return Value

[IDimXpertAnnotation](#)

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPart.InsertGtol](#).

## Remarks

Before calling this method, select the leader attachment points.

This method creates an empty Gtol symbol with a default value of 0.02 for Tolerance 1. To fill in the text and symbols, get the underlying SOLIDWORKS Gtol object and use [IGtol::SetFrameSymbols2](#) and [IGtol::SetFrameValues2](#) to change frame symbols and values.

## See Also

[IDimXpertPart Interface](#)

[IDimXpertPart Members](#)

## Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

**InsertLocationDimension Method (IDimXpertPart)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : InsertLocationDimension Method (IDimXpertPart)

Inserts a linear or angular location dimension between a selected feature and its locating feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Function InsertLocationDimension( _  
    ByVal Option As DimXpertDimensionOption _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Option As DimXpertDimensionOption  
Dim value As System.Boolean  
  
value = instance.InsertLocationDimension(Option)
```

C#

```
System.bool InsertLocationDimension(  
    DimXpertDimensionOption Option  
)
```

C++/CLI

```
System.bool InsertLocationDimension(  
& DimXpertDimensionOption^ Option  
)
```

#### Parameters

*Option*

[IDimXpertDimensionOption](#) or Null to dimension the location of the face along an axis normal to the locating feature and to place the dimension text at a default position

#### Return Value

True if location dimension is created, false if not

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart.InsertLocationDimension](#).

#### Example

[Get and Set Location Dimension \(C#\)](#)  
[Get and Set Location Dimension \(VB.NET\)](#)  
[Get and Set Location Dimension \(VBA\)](#)

#### Remarks

Before calling this method:

1. Call [IDimXpertPart::GetDimOption](#) to get an instance of IDimXpertDimensionOption.
2. For distance-between dimensions, set [IDimXpertDimensionOption::TextPosition](#) or [IDimXpertDimensionOption::DimensionPositionOption](#) (if both are set, only DimensionPositionOption is in force). For angle-between dimensions, set [IDimXpertDimensionOption::TextPosition](#).
3. Populate Option with the instance of IDimXpertDimensionOption.
4. Multi-select:
  - a. Origin Feature
  - b. Tolerance (locating) Feature

#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart::AutoDimensionScheme Method \(\)](#)  
[IDimXpertPart::InsertBasicDimension Method \(\)](#)  
[IDimXpertPart::InsertDatum Method \(\)](#)  
[IDimXpertPart::InsertPattern Method \(\)](#)  
[IDimXpertPart::InsertSizeDimension Method \(\)](#)

#### Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0



Inserts a linked, manual, or collection pattern using the selected faces.

## .NET Syntax

### Visual Basic (Declaration)

```
Function InsertPattern( _  
    ByVal Option As DimXpertDimensionOption, _  
    ByVal PatternType As System.Integer, _  
    ByVal FindAll As System.Boolean _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertPart  
Dim Option As DimXpertDimensionOption  
Dim PatternType As System.Integer  
Dim FindAll As System.Boolean  
Dim value As System.Boolean  
  
value = instance.InsertPattern(Option, PatternType, FindAll)
```

### C#

```
System.bool InsertPattern(  
    DimXpertDimensionOption Option,  
    System.int PatternType,  
    System.bool FindAll  
)
```

### C++/CLI

```
System.bool InsertPattern(  
& DimXpertDimensionOption^ Option,  
& System.int PatternType,  
& System.bool FindAll  
)
```

## Parameters

*Option*

[IDimXpertDimensionOption](#)

*PatternType*

0 = linked

1 = manual

2 = collection

*FindAll*

True to select similar faces for PatternTypes 1 and 2, false to not. (see **Remarks**)

## Return Value

False

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPart.InsertPattern](#).

## Example

[Get and Set Pattern \(C#\)](#)

[Get and Set Pattern \(VB.NET\)](#)

[Get and Set Pattern \(VBA\)](#)

## Remarks

Before calling this method:

1. Call [IDimXpertPart::GetDimOption](#) to get an instance of IDimXpertDimensionOption.
2. Set [IDimXpertDimensionOption::FeatureSelectorOptions](#) to swDimXpertFeatureSelectorOption\_Default to enable the FindAll property. If IDimXpertDimensionOption::FeatureSelectorOptions is set to anything other than the default, the FindAll setting is not in force.)

3. Populate Option with the instance of IDimXpertDimensionOption.
4. Select the features according to PatternType:
  - a. If PatternType is linked, then select one feature to automatically select all features of the pattern.
  - b. If PatternType is manual or collection and FindAll is set to true, then select one feature to automatically select all similar features on the face.
5. Set the selection mark for each selected feature to a unique value that is greater than 50.

#### See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart::AutoDimensionScheme Method \(\)](#)  
[IDimXpertPart::InsertBasicDimension Method \(\)](#)  
[IDimXpertPart::InsertDatum Method \(\)](#)  
[IDimXpertPart::InsertLocationDimension Method \(\)](#)  
[IDimXpertPart::InsertSizeDimension Method \(\)](#)

#### Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

#### InsertSizeDimension Method (IDimXpertPart)

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPart Interface](#) : InsertSizeDimension Method (IDimXpertPart)

Inserts a size dimension for the selected face or edge.

#### .NET Syntax

##### Visual Basic (Declaration)

```
Function InsertSizeDimension( _
    ByVal Option As DimXpertDimensionOption _
) As System.Boolean
```

##### Visual Basic (Usage)



```
Dim instance As IDimXpertPart
Dim Option As DimXpertDimensionOption
Dim value As System.Boolean

value = instance.InsertSizeDimension(Option)
```

##### C#

```
System.bool InsertSizeDimension(
    DimXpertDimensionOption Option
)
```

##### C++/CLI

```
System.bool InsertSizeDimension(
    & DimXpertDimensionOption^ Option
)
```

#### Parameters

*Option*

[IDimXpertDimensionOption](#)

#### Return Value

True if size dimension is created, false if not

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::InsertSizeDimension](#).

#### Example

[Get and Set Size Dimension \(C#\)](#)  
[Get and Set Size Dimension \(VB.NET\)](#)  
[Get and Set Size Dimension \(VBA\)](#)

## Remarks

Before calling this method:

1. Call [IDimXpertPart::GetDimOption](#) to get an instance of IDimXpertDimensionOption.
2. Set [IDimXpertDimensionOption::TextPosition](#) and/or [IDimXpertDimensionOption::FeatureSelectorOptions](#).
3. Populate Option with the instance of IDimXpertDimensionOption.
4. Select a face or edge.

## See Also

[IDimXpertPart Interface](#)  
[IDimXpertPart Members](#)  
[IDimXpertPart::AutoDimensionScheme Method \(\)](#)  
[IDimXpertPart::InsertBasicDimension Method \(\)](#)  
[IDimXpertPart::InsertDatum Method \(\)](#)  
[IDimXpertPart::InsertLocationDimension Method \(\)](#)  
[IDimXpertPart::InsertPattern Method \(\)](#)

## Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

ShowToleranceStatus Property (IDimXpertPart)

[See Also](#) [Example](#)



SolidWorks.Interop.swdimxpert Namespace > [IDimXpertPart Interface](#) : ShowToleranceStatus Property (IDimXpertPart)

Gets and sets whether tolerance status is displayed in the model.

## .NET Syntax

### Visual Basic (Declaration)

Property ShowToleranceStatus As System.Boolean

### Visual Basic (Usage)



```
Dim instance As IDimXpertPart
Dim value As System.Boolean

instance.ShowToleranceStatus = value
value = instance.ShowToleranceStatus
```

### C#

```
System.bool ShowToleranceStatus {get; set;}
```

### C++/CLI

```
property System.bool ShowToleranceStatus {
    System.bool get();
    void set ( & System.bool value);
}
```

### Property Value

True if tolerance status is displayed; false if not

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPart::ShowToleranceStatus](#).

## Example

[Auto Dimension Scheme \(C#\)](#)  
[Auto Dimension Scheme \(VB.NET\)](#)  
[Auto Dimension Scheme \(VBA\)](#)

## See Also

 **Availability**

SOLIDWORKS 2011 FCS, Revision Number 19.0

# IDimXpertPatternFeature (Type)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPatternFeature Interface

Allows you to access specific information about DimXpert patterns on the DimXpertManager tab of the Management Panel.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

## .NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertPatternFeature

Visual Basic (Usage)



Dim instance As [IDimXpertPatternFeature](#)

C#

public interface IDimXpertPatternFeature

C++/CLI

public interface class IDimXpertPatternFeature

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPatternFeature](#).

## Example

[Get and Set Pattern Example \(C#\)](#)  
[Get and Set Pattern Example \(VB.NET\)](#)  
[Get and Set Pattern Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

## Remarks

You can access more general DimXpert feature information using [IDimXpertFeature](#).

## Accessors

[IDimXpertPart::GetFeatures](#) and [IDimXpertPart::IGetFeatures](#)  
[IDimXpertPart::GetFeature](#)  
[IFeature::GetSpecificFeature2](#)

## See Also

[IDimXpertPatternFeature Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPatternFeature Interface

The following tables list the members exposed by [IDimXpertPatternFeature](#).

## Public Properties

| Name                        | Description                                     |
|-----------------------------|-------------------------------------------------|
| <a href="#">PatternType</a> | Gets the type of this DimXpert pattern feature. |

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| Public Methods                                                                                                       |                                                                    |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Name                                                                                                                 | Description                                                        |
|  <a href="#">GetSubFeatureCount</a> | Gets the number of sub-features for this DimXpert pattern feature. |
|  <a href="#">GetSubFeatures</a>     | Gets all of the sub-features of this DimXpert pattern feature.     |
|  <a href="#">IGetSubFeatures</a>    | Gets all of the sub-features of this DimXpert pattern feature.     |

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| See Also |
|----------|
|----------|

[IDimXpertPatternFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

IDimXpertPatternFeature Interface Methods

[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPatternFeature Interface

For a list of all members of this type, see [IDimXpertPatternFeature members](#).

| Public Methods                                                                                                       |                                                                    |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Name                                                                                                                 | Description                                                        |
|  <a href="#">GetSubFeatureCount</a> | Gets the number of sub-features for this DimXpert pattern feature. |
|  <a href="#">GetSubFeatures</a>     | Gets all of the sub-features of this DimXpert pattern feature.     |
|  <a href="#">IGetSubFeatures</a>    | Gets all of the sub-features of this DimXpert pattern feature.     |

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| See Also |
|----------|
|----------|

[IDimXpertPatternFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

IDimXpertPatternFeature Interface Properties

[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPatternFeature Interface

For a list of all members of this type, see [IDimXpertPatternFeature members](#).

| Public Properties                                                                                               |                                                 |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Name                                                                                                            | Description                                     |
|  <a href="#">PatternType</a> | Gets the type of this DimXpert pattern feature. |

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| See Also |
|----------|
|----------|

[IDimXpertPatternFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

GetSubFeatureCount Method (IDimXpertPatternFeature)

[See Also](#) [Example](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPatternFeature Interface](#) : GetSubFeatureCount Method (IDimXpertPatternFeature)  
Gets the number of sub-features for this DimXpert pattern feature.

| NET Syntax                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------|
| Visual Basic (Declaration)                                                                                           |
| Function GetSubFeatureCount() As System.Integer                                                                      |
| Visual Basic (Usage)  Copy Code |

```
Dim instance As IDimXpertPatternFeature
Dim value As System.Integer

value = instance.GetSubFeatureCount()
```

C#

```
System.int GetSubFeatureCount()
```

C++/CLI

```
System.int GetSubFeatureCount();
```

#### Return Value

Number of sub-features in the pattern

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPatternFeature::GetSubFeatureCount](#).

#### Example

[Get and Set Pattern Example \(C#\)](#)  
[Get and Set Pattern Example \(VB.NET\)](#)  
[Get and Set Pattern Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VB.NET\)](#)

#### See Also

[IDimXpertPatternFeature Interface](#)  
[IDimXpertPatternFeature Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

**GetSubFeatures Method (IDimXpertPatternFeature)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPatternFeature Interface](#) : GetSubFeatures Method (IDimXpertPatternFeature)

Gets all of the sub-features of this DimXpert pattern feature.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetSubFeatures() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPatternFeature
Dim value As System.Object

value = instance.GetSubFeatures()
```

C#

```
System.object GetSubFeatures()
```

C++/CLI

```
System.Object^ GetSubFeatures();
```

#### Return Value

Array of [IDimXpertFeatures](#)

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPatternFeature::GetSubFeatures](#).

## Example

[Get and Set Pattern Example \(C#\)](#)  
[Get and Set Pattern Example \(VB.NET\)](#)  
[Get and Set Pattern Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VB.NET\)](#)

## Remarks

This method returns all sub-features of the pattern feature. All of the sub-features returned by this method do not appear on the DimXpertManager tab of the Management Panel. Functional geometry sub-features such as planes and cylinders that represent each individual face of a part are for internal use only. These sub-features do not appear in the DimXpertManager tree. However, manufacturing sub-features that are significant for downstream machining and inspection of the pattern do appear in the DimXpertManager tree.

For example, the sub-features of a simple hole are related to functional geometry and are not listed in the DimXpertManager tree. However, the sub-features of a hole pattern are manufacturing features (e.g., simple holes) that are significant for downstream processes. These manufacturing features are listed in the DimXpertManager tree.

## See Also

[IDimXpertPatternFeature Interface](#)  
[IDimXpertPatternFeature Members](#)

## Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

**IGetSubFeatures Method (IDimXpertPatternFeature)**

[See Also](#)



SolidWorks.Interop.swdimxpert Namespace > [IDimXpertPatternFeature Interface](#) : IGetSubFeatures Method (IDimXpertPatternFeature)

Gets all of the sub-features of this DimXpert pattern feature.

## .NET Syntax

Visual Basic (Declaration)

```
Function IGetSubFeatures( _  
    ByVal Count As System.Integer _  
) As DimXpertFeature
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPatternFeature  
Dim Count As System.Integer  
Dim value As DimXpertFeature  
  
value = instance.IGetSubFeatures(Count)
```

C#

```
DimXpertFeature IGetSubFeatures(  
    System.int Count  
)
```

C++/CLI

```
DimXpertFeature^ IGetSubFeatures(  
& System.int Count  
)
```

### Parameters

*Count*

Number of sub-features in this DimXpert pattern feature

#### Return Value

- in-process, unmanaged C++: Pointer to an array of [IDimXpertFeature](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

#### Remarks

This method returns all sub-features of the pattern feature. All of the sub-features returned by this method do not appear on the DimXpertManager tab of the Management Panel. Functional geometry sub-features such as planes and cylinders that represent each individual face of a part are for internal use only. These sub-features do not appear in the DimXpertManager tree. However, manufacturing sub-features that are significant for downstream machining and inspection of the pattern do appear in the DimXpertManager tree.

For example, the sub-features of a simple hole are related to functional geometry and are not listed in the DimXpertManager tree. However, the sub-features of a hole pattern are manufacturing features (e.g., simple holes) that are significant for downstream processes. These manufacturing features are listed in the DimXpertManager tree.

See [In-process Methods](#) for details about this type of method.

Before calling this method, call [IDimXpertPatternFeature::GetSubFeatureCount](#) to get the value for the Count parameter.

#### See Also

[IDimXpertPatternFeature Interface](#)  
[IDimXpertPatternFeature Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

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**PatternType Property (IDimXpertPatternFeature)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPatternFeature Interface](#) : PatternType Property (IDimXpertPatternFeature)

Gets the type of this DimXpert pattern feature.

#### .NET Syntax

##### Visual Basic (Declaration)

ReadOnly Property PatternType As [swDimXpertFeatureType\\_e](#)

##### Visual Basic (Usage)



```
Dim instance As IDimXpertPatternFeature  
Dim value As swDimXpertFeatureType\_e  
  
value = instance.PatternType
```

##### C#

```
swDimXpertFeatureType\_e PatternType {get;}
```

##### C++/CLI

```
property swDimXpertFeatureType\_e PatternType {  
    swDimXpertFeatureType\_e get();  
}
```

#### Property Value

Type of this DimXpert pattern feature as defined in [swDimXpertFeatureType\\_e](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPatternFeature::PatternType](#).

#### Example

[Get and Set Pattern Example \(C#\)](#)  
[Get and Set Pattern Example \(VB.NET\)](#)  
[Get and Set Pattern Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VB.NET\)](#)

#### See Also

[IDimXpertPatternFeature Interface](#)  
[IDimXpertPatternFeature Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0